D

Dampeners (*see* Pulsation Dampeners)

Desalination

How Desalination Works

There are many different techniques for desalination. Among the main ones are the multistage flash (MSF) process and reverse osmosis. Countries that do not have an abundance of fresh water are prime candidates for the use of this technology. Quatar's installations include the Ras Abu Fontas B power and desalination plant, which cost $1 billion to build. Dubai in the United Arab Emirates (UAE) has a 60-million-gal/day desalination plant at Jebel Ali. The plant's eight MSF units are part of a cogeneration power facility. The fresh water would have been at least as much incentive in the Middle East as the increased total thermal efficiency. The desalination equipment in both cases was supplied by Weir Westgarth (WW).

WW designed the MSF process. The principle of the system is simple: Water and steam in a closed system can be made to boil at temperatures lower than at standard temperature and pressure by reduction of the system pressure. MSF plants contain a series of closed chambers—as many as 20—each held at a lower pressure than the preceding one.

Heated salt water is passed through the overall system. Some of the salt water in each chamber vaporizes into steam. Moisture droplet separators remove salt water droplets. The steam condenses to fresh water when faced with cold tubes and is collected for storage. The last chamber's brine is quite cool, and it is, in fact, used as the coolant fluid. Then it starts to pick up the latent heat of condensation and increases in temperature. Only a small amount of additional heat is required to prepare this steam for entry into the first flash chamber. One source of this steam is low-pressure steam from a power station.

The key to the reverse osmosis (RO) process is a suitable semipermeable membrane. Improvements in membrane technology now mean that the process can apply to industrial-scale plants. Common contemporary membrane selections are made of cellulose-based polymer or a polyamide layer applied to a microporous polymer film. This membrane is bonded to a porous polyester sheet for structural stiffness. This composite is rolled into a spiral. Spun hollow fine fibers are the finished product. The semipermeable layer is on the outside of the fibers. The total thickness of the composite is about 24 μm. The outside diameter of the tube is about 95 μm, making for a large surface area for rejecting salt. The fibers are made into bundles that are sealed with epoxy in a fiberglass pressure container.

The Global Drive for Desalination

The motivation level for incorporating desalination plants into plant infrastructure varies. The Middle East has fuel resources in abundance—natural gas, clean oil, and residual oil—and technology available that will help it burn each of these options with acceptable efficiency. Desalination technology also helps the Middle East achieve its major operational objective, which is to extend the time between overhauls (TBOs) and therefore parts life [as defined by life-cycle analysis (LCA)],

over the previous generation of power-producing machinery operated. Saving fuel for its own sake is not as critical to this market as it is to areas that do not have natural gas and oil in abundance. However, the Middle East has another reason for caring about efficiency—other than TBOs and LCA—that has emerged in the last decade: desalination. The Middle East is very short of fresh water.

The main forces behind the optimized commercialization of desalination technology are ironies in light of conditions in the Middle East. Concern regarding seawater contamination of freshwater aquifers in more freshwater-rich countries was one. (Sea water takes the place of fresh water as the aquifer pressure drops with water extraction.) The need to give nuclear energy a better image (by using its waste heat to produce fresh water) was another.

With respect to the first concern: desalination alleviates use of underground freshwater aquifers, which then reduces the risk of sea water seeping in to make up the balance of fresh water pumped out of the ground. Japan is gas and oil poor. A major user of nuclear power, it has been instrumental in promoting the use of desalination processes in conjunction with its nuclear facilities. The water is produced only in enough quantities to be used by the plants themselves. However, the Nuclear Power Technology Development section of the International Atomic Energy Agency (IAEA) has been seeking to promote large-scale desalination in conjunction with nuclear power production since 1989. The IAEA no doubt hopes that a freshwater supply would overcome the general public's reluctance to be situated close to a nuclear station. Water rises in cost if it has to be transported further.

Interestingly, despite the IAEA agreeing that MSF is promising, they are not contemplating using it for any large-scale attempts at freshwater production because of its inflexibility at partial load operation. MSF's tendency to corrosion and scaling versus other desalination techniques is also a factor.

Material Metallurgical Selections

The amount of sea water handled for a given membrane selection and seawater temperature varies directly with the applied pressure. Gulf sea water typically contains 19,000 ppm of salt. Typical temperature gradients are 20 to 30°C. Metallic corrosion at 30°C is 4 times what it would be at 20°C (double the rate of corrosion for each 5°C rise in temperature). This then requires corrosion-resistant alloy selections for the pumps.

In a design for a 25-year pump life for a Danish nuclear plant, Alfa Laval used copper nickel alloys in the evaporator and titanium for the heat-transfer tubing.

Desalinated water requires fewer chemical additives for water treatment. This is an advantage in terms of overall system cost.

Dialysis; Electrodialysis

Dialysis or *electrodialysis* is a specialized chemical process to separate components using differences in diffusion rates through membranes. A difference in concentrations drives the flow of the components in question. This process is used in chemical purification systems.

Distillation; Fractional Distillation (*see also* Towers and Columns)

Distillation is the evaporation and subsequent collection of a liquid that is a component in a mixture. *Fractional distillation* is the evaporation of two or more

liquids from a parent mixture by using the differences in their boiling points. Both processes may be used for purification or separation. A typical example of distillation is extracting pure solvent (such as water) from a mixture of solute and solvent (such as brine).

A typical example of fractional distillation is in the separation of various hydrocarbons, for instance, butane, pentane from a hydrocarbon mixture that results from some process in the overall refining process. In this example, a refinery fractional distillation apparatus is commonly termed a *fractional distillation column* or a *fractionating column* and can be several hundred feet high, depending on the refinery throughput. This column then consists basically of an outer shell, several metallic trays arranged through the column's height to enable the drawing off of various liquids. Pumps and pipes inside and outside the column conduct the fluid in required directions. Pressure and temperature in the column is controlled to the most suitable values for the process in question. Degrees of vacuum are used to accelerate separation rates.

A vessel of this nature is custom designed, generally by the design contractor for the overall plant. Besides all the relevant physical property tables, designers will use past experience to a considerable extent to determine the final parameters and dimensions for the column. In prototype applications some estimation or guesswork is unavoidable, which may have consequences not necessarily within the column itself, but with higher-precision elements of the process downstream. For example, a fractionating column-handling process oil in an oil sands plant had various molecular weights of hydrocarbon taken off at various points along its length. Based on flash point, one of these streams was designated as the source supply for the purged seal oil system in the plant. That stream contained more colloidal coke than was expected, making it unsuitable for the application. The problem was solved by taking the required supply from higher up the column (a "lighter end" of lower molecular weight).

Diverter; Diverter Damper; Diverter Valve; Flapper Valve

Diverter commonly describes a flat-plate-type valve hinged at one end that is moved to divert flow from one stack or passage to the other. There is normally no appreciable degree of speed or mobility attached to this type of device, as there might be with a butterfly valve in a control system. In a 150,000-barrel/day refinery, for instance, a flapper valve or flapper diverter valve can be made of concrete and weigh up to 5 tons. The absolute closure of the closure seal may not be critical, and the allowable gap between the flapper surface and the mating flange it sits on may be as large as 0.010 in.

Doctor

A *doctor* is a paper industry device. It is used to lead the paper sheet and keep the paper roll(s) clean.

Drives (*see also* Power Transmission; Turbines)

Drives is the term given to power-transmission equipment. The simplest form of power transmission is a belt drive. A belt drive can have a flat, V-belt, ribbed belt, or toothed belt design. These drives are common in applications such as conveyors. Conveyor manufacturers should be consulted for their catalogues on power-transmission capability.

In modern process industry, drive equipment might be classified as power-transmission equipment.

Drum; Knock-Out Drum; Knock-Out Vessel (*see* Separators)

Drying

These are many types of drying media and methods. A variety of methods to produce heat for drying (ovens) or air for drying (fans) or moisture absorption chemical (desiccant) are used and generally customized for a specific application. There are fluid-bed dryers, where solid particulates disperse heat to a gas. Sophistication may be added by using different stages for the dryer with dust collection (cyclone separators or otherwise) at each phase. In a pneumatic conveying system, an agitator supplied with warm air can be used to accelerate drying. With vacuum dryers, moisture removal is completed below atmospheric pressure. Drying temperatures, however, may vary considerably. Vacuum drying is particularly popular in pharmaceutical manufacturing, electronics, metallurgical, and food industries. *See* Some Commonly Used Specifications, Codes, Standards, and Texts.

Drying Equipment; Driers

The simplest kind of drying is "open-air" drying. From there, we can progress to fans, heaters, hot-air blowers, ovens, conveyor and oven systems, heat-exchanger provided heat, and a variety of other options, too numerous to concentrate on extensively in this book.

Drying, Freeze*

Freeze drying is sometimes what is meant by the term "drying" in the process engineer's terms. The following information is based on the solvent Tebol 99. It indicates what freeze drying is and the properties sought and the performance parameters measured in a typical freeze-drying agent.

Freeze drying or lyophilization is a process that removes a solvent, typically water, from a frozen solution by sublimation. Studies in the 1930s and 1940s were done on blood serum and foods. More recently, research has focused on using freeze drying for pharmaceuticals, cosmetics, and chemicals. An increasing number of parenteral products have been prepared by freeze-drying techniques. The method reduces particulate contamination, improves product quality and stability, and enhances the dissolution rate on reconstitution.

The pharmaceutical industry takes advantage of the freeze-drying process to maintain the activity and viability of various delicate biological materials. These materials include antibiotics, peptides, proteins, vaccines, and microbial cells. While freeze drying with water has proven useful, it has several inherent limitations:

- Uneven moisture distribution in the freeze-dried product
- Uneven stability or unpredictability of the final product
- Useless for water-insoluble or hydrolyzable products
- High energy costs
- Long process cycles

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*Source: ARCO Chemical, USA.

Perhaps most important, freeze drying with water is restricted to those materials that are soluble and stable in a water system.

Much attention has been devoted to optimizing freeze-drying cycles. Recent studies have shown that addition of tertiary butyl alcohol (TBA) can markedly improve the freeze-drying process.

TBA as a processing aid:

- Helps dissolve products that are difficult to dissolve in water
- Gives a product with a high specific surface area
- Accelerates the drying process by reducing dried product resistance
- Prevents the product from reaching the collapse temperature
- Produces a pharmaceutically elegant product that can be reconsitituted easily

Typical physical properties related to freeze drying (see Table D-1)

TBA-water phase diagram. Figure D-1 shows the phase diagram for the TBA/water system developed by Kasraian and DeLuca. Water and TBA form a TBA hydrate. This complex phase diagram essentially consists of two simple eutectic phase diagrams placed side by side. The left side represents the eutectic phase diagram for water–TBA hydrate; the right side represents that of TBA hydrate–TBA. The maximum at 70 percent TBA corresponds to the melting of the pure TBA hydrate. The TBA-water system has two eutectic compositions, one at 20 percent TBA (eutectic A) and the other at 90 percent TBA (eutectic B). For the purpose of accelerating the freeze-drying process, only 5–10 percent of TBA is needed. In the 5–10 percent TBA concentration range, TBA and water form a eutectic mixture that has a melting point of –5 °C. Therefore, during freeze drying the product temperature should be kept below –5 °C.

Rates of sublimation of TBA/water solution. The relative rates of sublimation for each component depend on its concentration. Figure D-2 compares sublimation rates by plotting the molar ratio of TBA remaining to water remaining versus time. During the sublimation of a 20 percent TBA solution, the TBA/water ratio remains constant, which shows that both sublime at the same rate. Higher TBA concentration solutions have negative slopes, indicating that TBA is subliming faster than ice. Lower TBA concentration solutions show that ice was subliming faster than TBA. Figures D-3 and D-4 show appropriate clothing for operator handling of this product.

TABLE D-1 Typical Properties

1. Flash point (tag closed cup), °C (°F)	11 (52)
2. Vapor pressure at 26 °C (mmHg)	46
3. Density (lb/gal at 26 °C)	6.5
4. Viscosity (cps) at 30 °C	3.3
5. Surface tension (dynes/cm) at 25 °C	19.6
6. Refractive index at 25 °C	1.38
7. Solubility parameter at 26 °C	
Hansen D	7.2
Hansen P	2.6
Hansen H	7.1
Total Hansen	10.4
8. Solubility, in H_2O at 26 °C	Complete

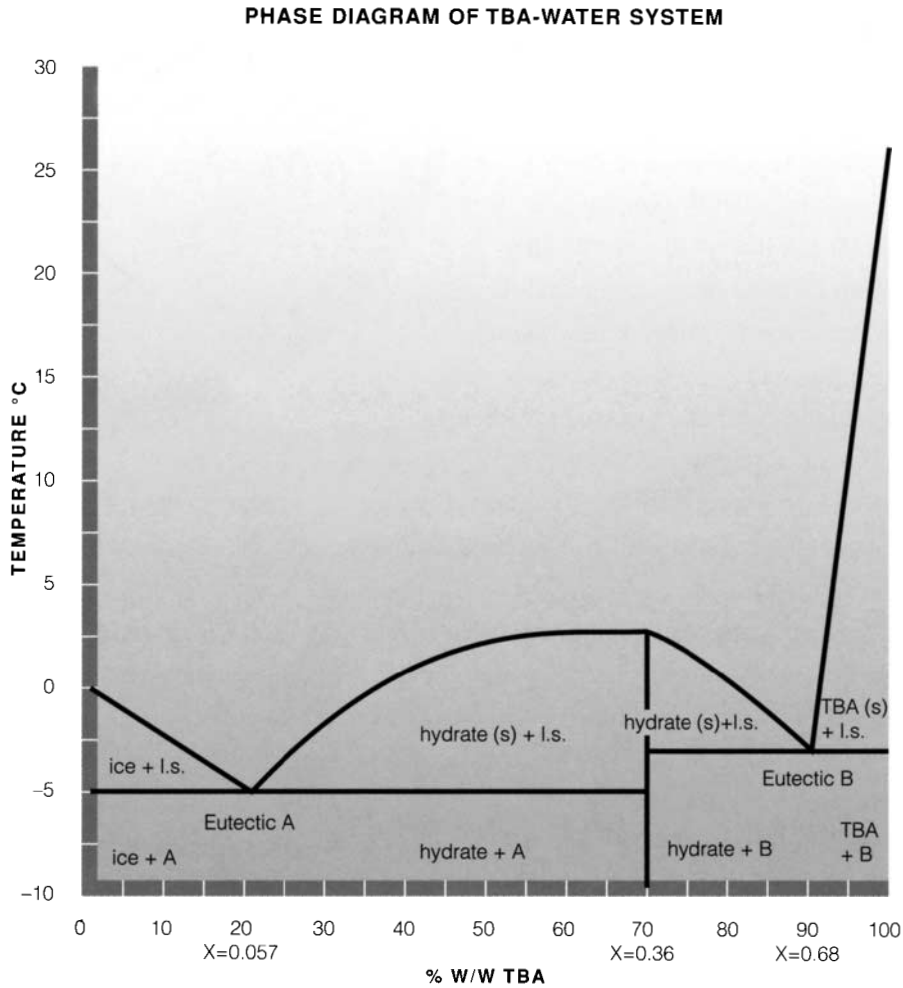


FIG. D-1 Phase diagram for TBA-water system. (l.s. = liq. state.) (Kasraian, K., DeLuca, P., *Pharmaceutical Research*, Vol. 12, No. 4, 1995; permitted by the Plenum Publishing Corporation.)

Frozen TBA-water mixtures. Research using freeze-drying microscopy has shown that TBA affects the crystal habit of ice and therefore the sublimation rate. Adding 3–19 percent TBA resulted in the formation of large needle-shaped ice crystal patterns that can facilitate sublimation. Once these crystals sublime, they leave behind a more porous and lower resistance dry matrix than water alone. Drying can take place more effectively through this matrix. Figure D-5 shows frozen TBA-water mixtures with TBA concentrations varying from 0 to 70 percent w/w. As TBA concentrations increase, the crystal patterns become more ordered and needle-shaped.

Porous resistance of the dry product layer. Research has shown that adding 5% w/v TBA into a 5 percent w/v sucrose solution considerably shortens the primary drying stage by lowering the resistance of the dried cake. Figure D-6 shows that the frozen solution without TBA initially had a high resistance, approximately 60 cm^2 torr hr/gm, due to the formation of a skin. Once the skin cracked, the resistance improved to 10 cm^2 torr hr/gm. The solution containing TBA had a resistance of 0.5–3 cm^2 torr hr/gm. Figure D-7 shows the result. Without TBA, the sucrose solution dried in 100 hours. Adding 5 percent w/v TBA lowered the drying time to 10 hours.

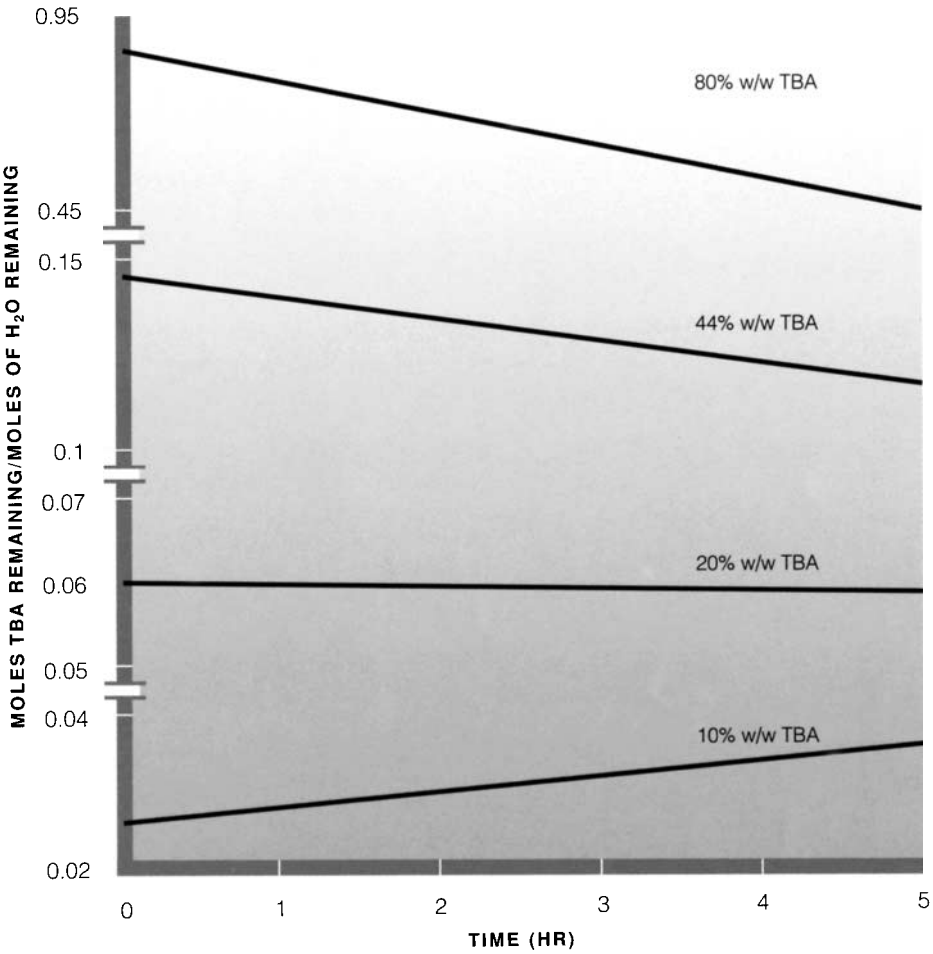


FIG. D-2 Ratio of TBA to water as a function of time for 10%, 20%, 44%, and 80% TBA solutions. (Source: Kasraian, K., DeLuca, P., *Pharmaceutical Research*, Vol. 12, No. 4, 1995; permitted by the Plenum Publishing Corporation.)



FIG. D-3 Suitable clothing for operator handling. (Source: ARCO Chemical.)



FIG. D-4 Suitable clothing for operator handling. (Source: ARCO Chemical.)

Applications

TBA as mass transfer accelerator. Beecham Pharmaceuticals has extensively studied the effect of organic solvents, especially TBA, on freeze-drying efficiency and product properties. (See Figs. D-8 and D-9.)

The use of TBA for freeze drying the common antibiotic gentamicin, in the presence of maltose, has been reported. Adding TBA reduced the drying time from 39 hours to 28 hours and maintained the porous structure of the product.

TBA in biopharmaceuticals. The industry uses sugar and other polyhydroxy compounds as stabilizers for proteins and biological materials in their formulations. Sugars are added to solutions for freeze drying to protect certain protein compounds from freeze, freeze-thaw, and freeze-drying damage. However, freeze-drying cycles of such solutions are excessively long because sugar solutions collapse at very low temperatures. Consequently, low shelf temperatures must be maintained throughout the drying stage.

The effect of adding 5 percent cosolvent on the freeze drying of sucrose and lactose has been extensively studied. Table D-2 compares the freeze-drying performance of organic solvents with water. All solutions, except the TBA ones, failed to freeze dry. The solvent systems without TBA experienced severe bubbling of the cosolvents followed by collapse during the drying phase. Solutions containing TBA resulted in complete drying and yielded good cakes.

The freeze-drying behavior of sugar solutions at various temperatures has been researched. It was shown that adding 5–10 percent w/v TBA increased the drying rate by 3 times (Table D-3). In addition, only the TBA-containing solutions survived at a 30°C shelf temperature.

The effects of using TBA on the properties of the dried sucrose (Table D-4) has been compared. Data indicate that the cake dried from the TBA solution was very porous. The hypothesis is that the porous nature causes reduced resistance to water vapor transfer during sublimation and subsequently gives faster drying rates.

Figures D-10 and D-11 provide scanning electron microscopy (SEM) and photographs of sucrose solutions during freeze drying without and with TBA. Collapse occurs in the samples without TBA. Therefore TBA appears to either

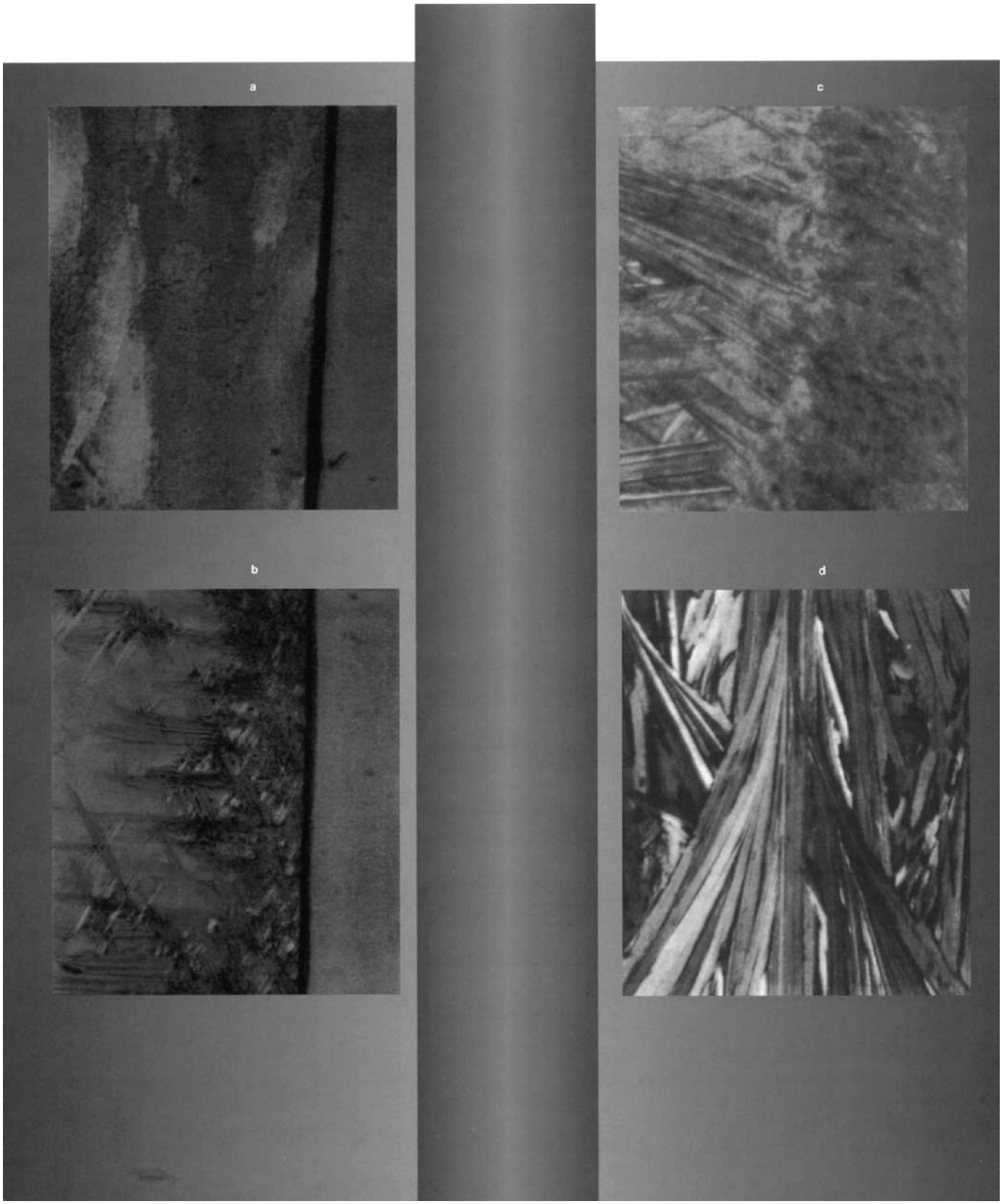


FIG. D-5 Polaroid photographs of frozen TBA-water mixtures with different concentrations of TBA: (*a*) frozen deionized water, (*b*) 10% w/w TBA aqueous solution, (*c*) 50% w/w TBA aqueous solution, (*d*) 70% w/w TBA aqueous solution. (Source: Kasraian, K., DeLuca, P., *Pharmaceutical Research*, Vol. 12, No. 4, 1995; permitted by the Plenum Publishing Corporation.)

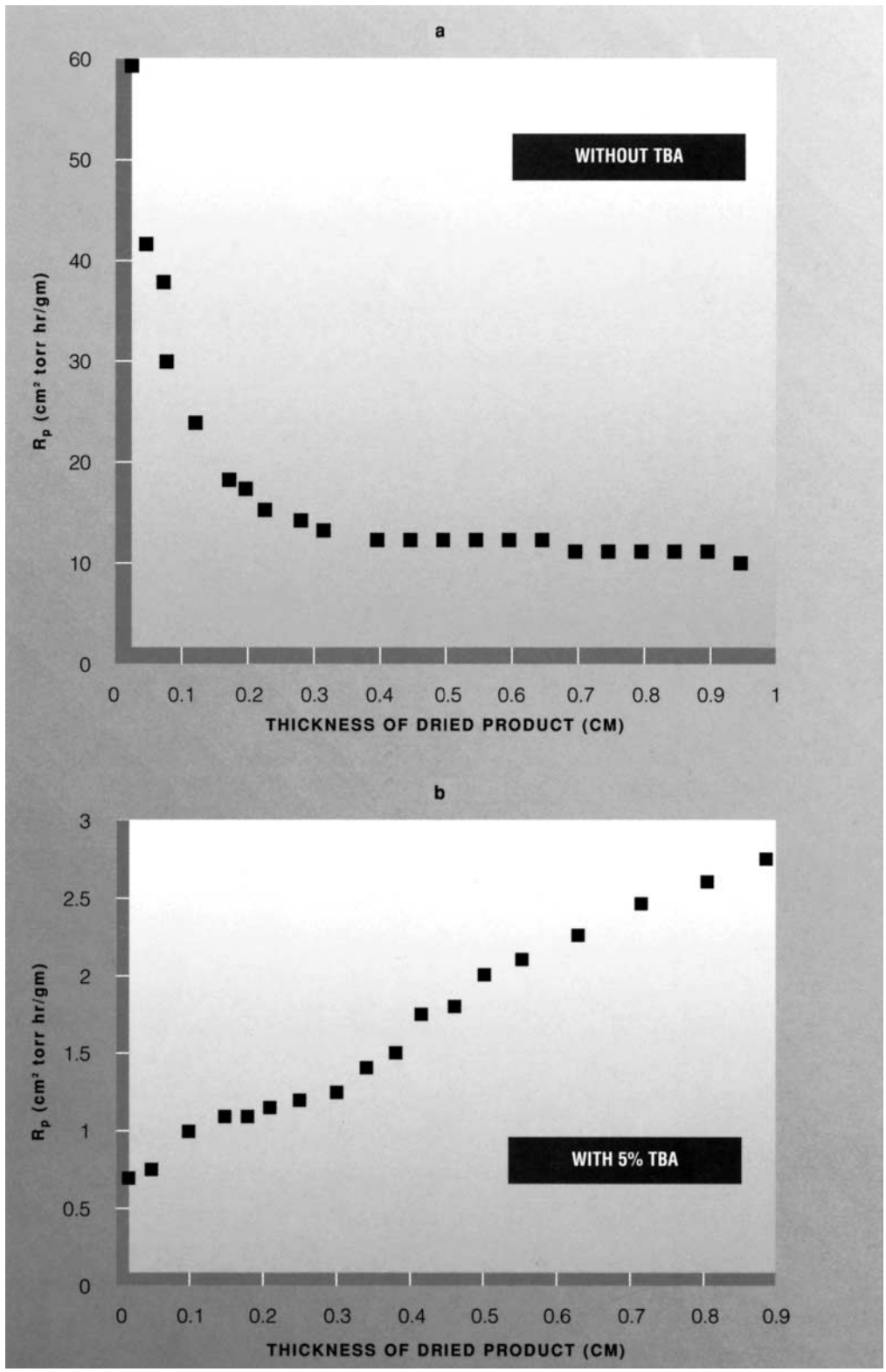


FIG. D-6 Normalized dried product resistance versus thickness of dried product: (*a*) 5% w/v sucrose freeze dried in a microbalance at a temperature of −35 °C, (*b*) 5% w/v sucrose containing 5% w/v TBA freeze dried in a microbalance at −35 °C. (Source: Kastaian, K., DeLuca, P., *Pharmaceutical Research*, Vol. 12, No. 4, 1995; permitted by the Plenum Publishing Corporation.)

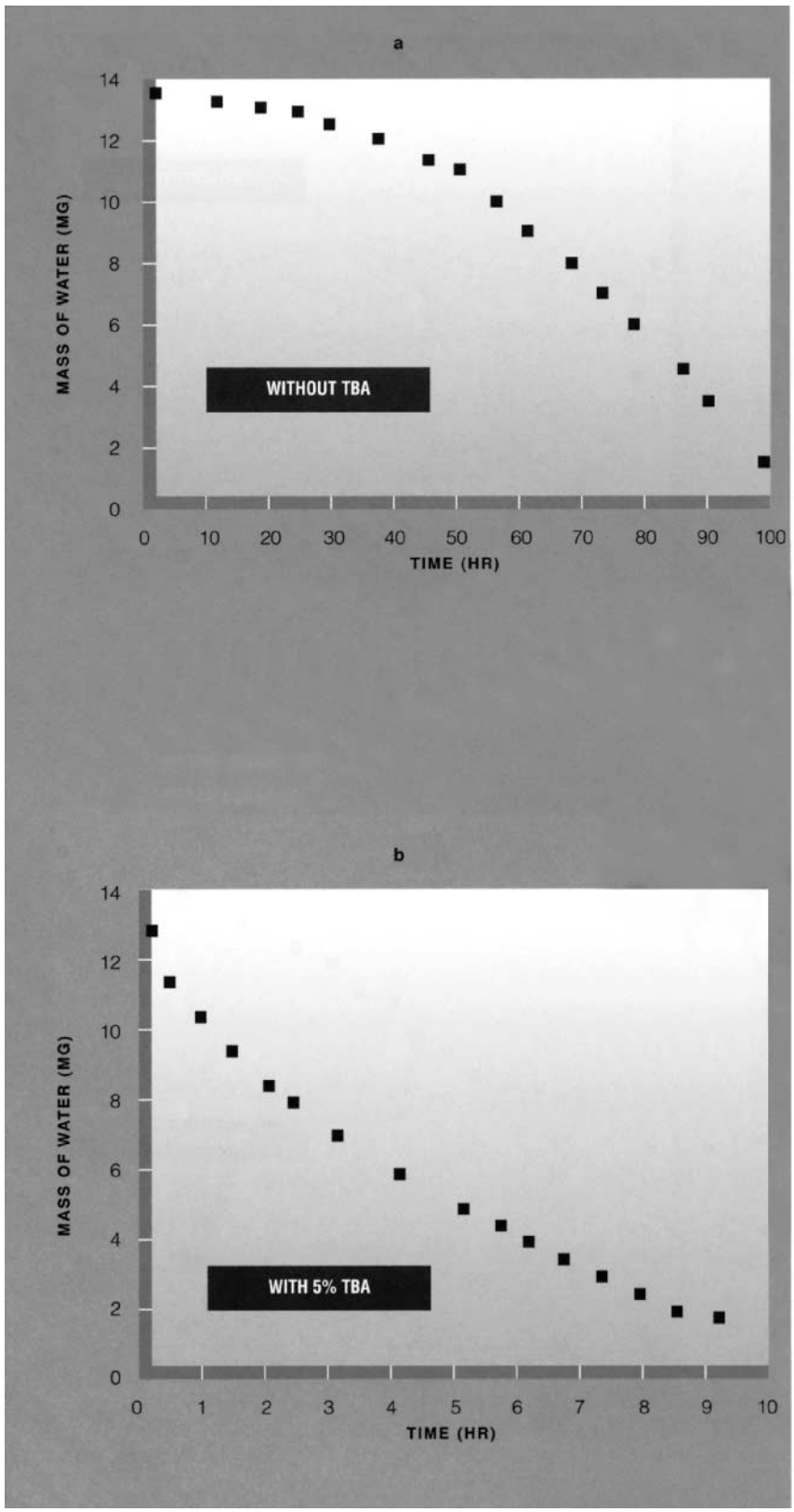




FIG. D-7 Mass loss of water from: (*a*) 5% w/v sucrose solution, (*b*) 5% w/v sucrose solution containing 5% w/v TBA. Drying temperature −35 °C in the microbalance. (Source: Kasraian, K., DeLuca, P., *Pharmaceutical Research*, Vol. 12, No. 4, 1995; permitted by the Plenum Publishing Corporation.)



FIG. D-8 A vacuum dryer. (Source: Stokes Vacuum Inc.)

elevate the collapse temperature or prevent the frozen product from reaching the collapse temperature due to faster rate of sublimation.

AKZO Pharma Division of Organon International B.V. also reported how TBA concentrations affect the stability of freeze-dried sucrose formulations. Adding 5 percent TBA to a 180 mg/ml sucrose solution resulted in a pharmaceutically acceptable, stable freeze-dried cake with no collapse. Therefore, while TBA addition did not change the collapse temperature of the sucrose solution, it increased the rate of sublimation. The increase thereby prevented the product from ever rising to the collapse temperature.

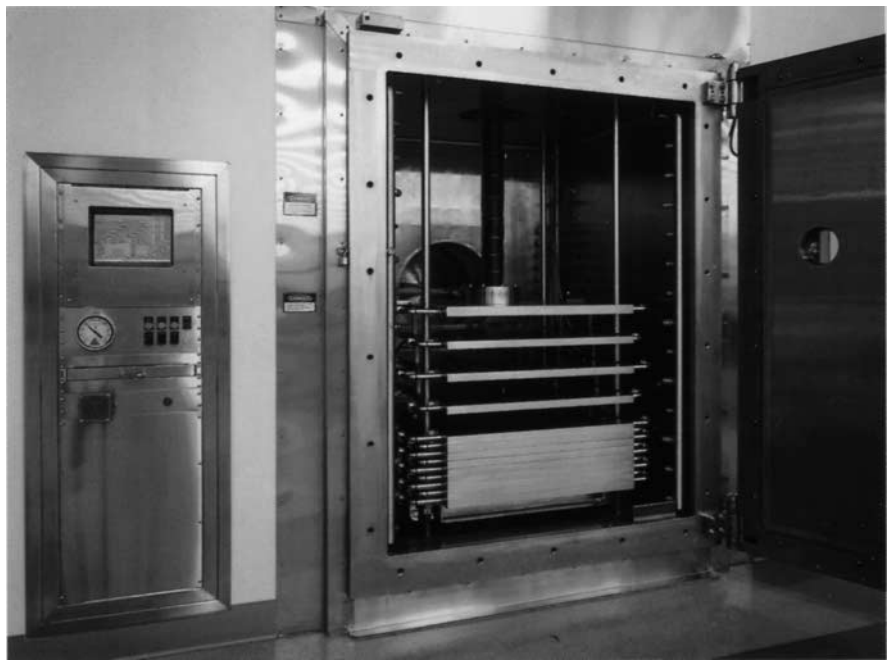


FIG. D-9 A vacuum dryer. (Source: Stokes Vacuum Inc.)

TABLE D-2 Effects of Solvent on Freeze-Drying Sugar Solutions

System	Shelf Temperature (°)	Remarks
Sucrose alone	0	Collapse
Sucrose + methanol	0	Boiling of solvent, collapse
Sucrose + ethanol	0	Boiling of solvent, collapse
Sucrose + isopropanol	0	Boiling of solvent, collapse
Sucrose + acetone	0	Boiling of solvent, collapse
Sucrose + *n*-butanol	0	Boiling of solvent, collapse
Sucrose + dioxane	0	Boiling of solvent, collapse
Sucrose + TBA	30, 45	Good
Lactose alone	0	Collapse
Lactose + methanol	0	Boiling of solvent, collapse
Lactose + ethanol	0	Boiling of solvent, collapse
Lactose + isopropanol	0	Boiling of solvent, collapse
Lactose + acetone	0	Boiling of solvent, collapse
Lactose + n-butanol	0	Boiling of solvent, collapse
Lactose + dioxane	0	Boiling of solvent, collapse
Lactose + TBA	30, 45	Good

Taken from DeLuca, P. P., Kamat, M. S., Koida, Y. Congr. Int. Technol. Pharm., 5th, 1989, 1, 439, permitted by the publisher, Rue J.-B. Clement.

TBA in freeze drying of lipids and liposomes. Use of liposomal formulations is rapidly gaining popularity in pharmaceutical research and development. Liposomes are increasingly serving as carriers for antigens and/or drugs for different routes of administration. The physical stability of the liposomes during long-term storage has been a matter of intense investigation for some time. To ensure its therapeutic properties, a liposomal dosage form must be stable with respect to its drug-carrier characteristics. While several approaches can stabilize liposomes, lyophilization is one of the best available methods to extend shelf-life.

TABLE D-3 Observations during Freeze-Drying Sugar Solutions (10% w/w)

	Freezing Pattern	Collapse Temperature, °C	Shelf Temperature, °C	Product Temperature, °C	Drying Rate, g/hr
Sucrose alone	Spontaneous	–28	–30	–31	0.21
Sucrose + 5% TBA	Slow, needles	–21	+30	–28	0.60
Sucrose + 10% TBA	Slow, needles	–20	+30	–30	0.67
Lactose alone	Spontaneous	–23	–15	–27	0.19
Lactose + 5% TBA	Slow, needles	–22	+30	–26	0.70
Lactose + 10% TBA	Slow, needles	–21	+30	–27	0.79

Taken from DeLuca, P. P., Kamat, M. S., Koida, Y. Congr. Int. Technol. Pharm., 5th, 1989, 1, 439, permitted by the publisher, Rue J.-B. Clement.

TABLE D-4 Effects of TBA on Properties of Dried Sucrose

	Without TBA	With TBA
Initial drying	Slow	Fast
Texture	Coarse	Smooth, soft
Particle shape	Irregular	Irregular
SEM	Plate-like	Plates but porous
Polarized light	Non-birefringent	Partial birefringence
Crystallinity	Amorphous	Partial crystallinity
Residual moisture	1–2%	1–2%
Surface area	0.9 m^2/g	1% TBA = 1.13 m^2/g 5% TBA = 2.25 m^2/g 10% TBA = 2.80 m^2/g
Residual TBA		<0.2%
Total drying time	40 hr	22 hr

Taken from DeLuca, P. P., Kamat, M. S., Koida, Y. Congr. Int. Technol. Pharm., 5th, 1989, 1, 439, permitted by the publisher, Rue J.-B. Clement.

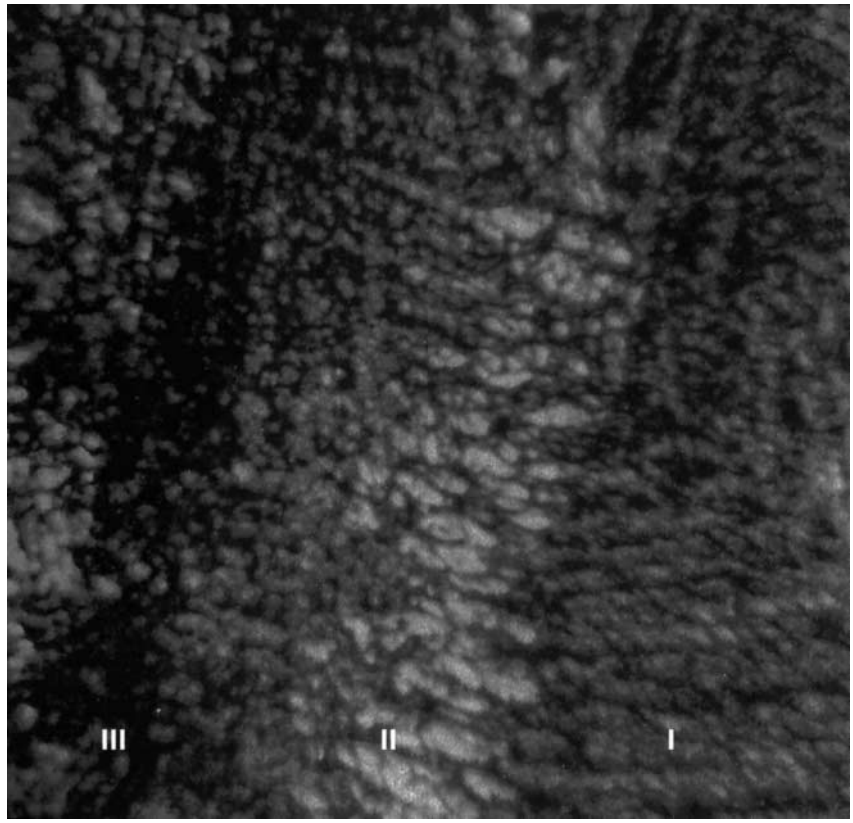
Because water is an essential component of the liposomal bilayer structure, freezing can promote damage due to dehydration. Use of cryoprotectants and control of the freezing rate that can minimize the formation of large ice crystals are of utmost importance. To achieve stability, the quality of phospholipids and the production process must be reproducible.

Research has shown that lipid solubility is four to five times greater in TBA than in other organic solvents such as ethanol. Therefore, more researchers are considering TBA as a lyophilization solvent for dissolution of lipids.

Ciba-Geigy Ltd., reported using TBA and *N*-methyl pyrrolidone (NMP) as water miscible organic solvents in large-scale production of liposomes. TBA was selected to dissolve the phospholipids and NMP to dissolve the dye, zinc phthalocyanine. This organic phase was mixed with an excess of a water phase to yield reproducible unilamellar liposomes with a mean size of 50–150 nm. The liposomes were then sterile filtered and freeze dried in a mixture of lactose and phospholipid. Three batches were tested for particle size, monomeric ZnPc, residual organic solvent, and moisture content in the lyophilized samples. Particle size was comparable after every manufacturing step with all three batches. The fraction of monomeric ZnPc was 100 percent in all three batches after every manufacturing step. Also, the removal of organic solvent and moisture content were reproducible.

Shionogi & Co., Ltd., reported a process for manufacturing a crystalline, lyophilized formulation of fosfomycin sodium (FOS) using aqueous TBA. FOS has an extremely high affinity with water, and the eutectic point is below –40 °C. An aqueous solution of FOS cannot be frozen at the temperatures obtained in common

(a)



(b)

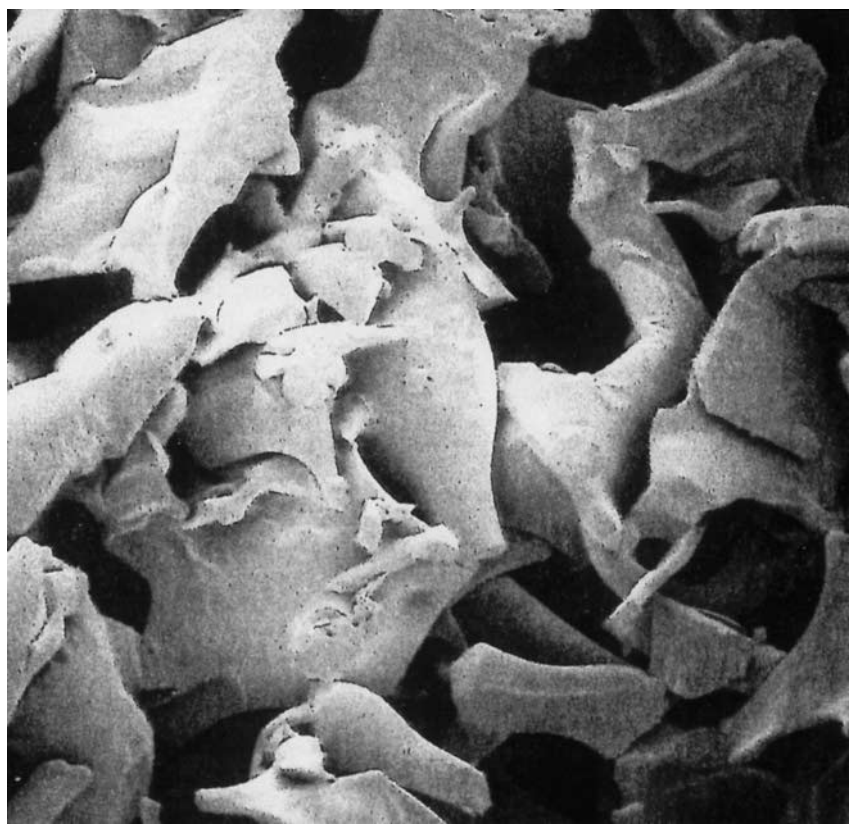
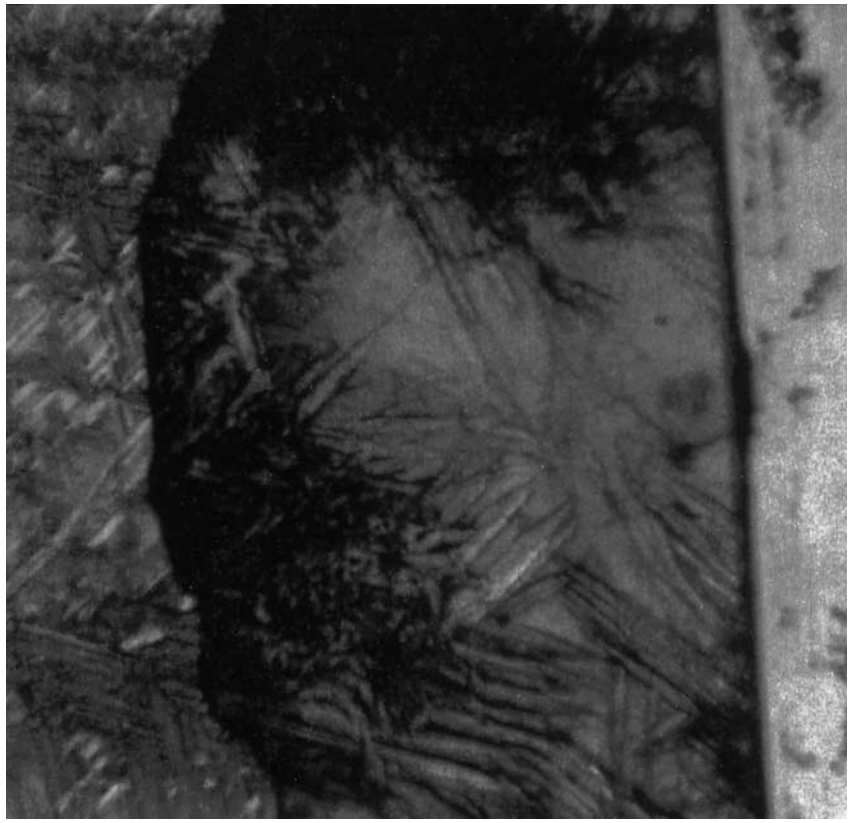


FIG. D-10 (*a*) Polaroid photographs of sucrose during freeze drying without TBA. Region I: dried material. Region II: collapse. Region III: frozen matrix. (*b*) SEM of freeze-dried sucrose (10% w/v). (Source: Kasraian, K., DeLuca, P., *Pharmaceutical Research*, Vol. 12, No. 4, 1995; permitted by the Plenum Publishing Corporation.)

(**a**)



(**b**)



FIG. D-11 (*a*) Polaroid photographs of sucrose during freeze drying in the presence of TBA. (*b*) SEM of sucrose (10% w/v) freeze dried with TBA. (Source: Kasraian, K., DeLuca, P., *Pharmaceutical Research*, Vol. 12, No. 4, 1995; permitted by the Plenum Publishing Corporation.)

freeze-drying devices. Moreover, the sample immediately melts during primary drying. For these reasons, no such formulations are presently available. Shionogi & Co., Ltd., developed a simple manufacturing method that produces a stable product with a long shelf-life. They have obtained the target formulations by dissolving FOS in aqueous TBA and then freeze drying. The TBA allows FOS to be freeze-dried at a temperature far higher than the original eutectic point. Therefore, ordinary freeze-drying equipment and operation can be used.

Sumitomo Pharmaceuticals Co., Ltd., reported a method for preparing a lyophilized formulation of a liposoluble platinum (II) complex. Liposoluble platinum (II) is virtually insoluble in water, but very soluble in TBA. Therefore, researchers dissolved the liposoluble complex in TBA and then lyophilized it for use as an anticancer drug. Analysis revealed that the resulting TBA content was no more than 0.05 wt%.

Many more patents and publications describe the use of TBA for dissolving liposomes. Geo-Centers, Inc., has patented a process for fabricating lipid microstructures using TBA where the dissolved lipid grows into tubular microstructures. Mehta et al. reported using TBA to lyophilize antifungal polyene macrolide-containing liposomes.

Freeze drying of water unstable drugs. The Upjohn Company reported a process to manufacture a stable, lyophilized formulation of prostaglandin E1 (PGE-1) for use in the treatment of erectile dysfunction. Lyophilization of a buffered lactose formulation of PGE-1 from a TBA/water mixture provides superior product stability than when freeze drying from a 100 percent aqueous system. The level of TBA that afforded the product maximum stability appeared to be when the TBA amount ranged from 17–25 percent (v/v). The unique kinetics of the degradation pathway of PGE-1 indicates that it is imperative to keep PGE-1 molecules as far apart as possible in order to minimize the interaction of two PGE-1 molecules. TBA is most likely enabling the PGE-1 molecules to be kept further apart during the freezing and lyophilization phases of manufacture.

Bristol-Myers Company has reported on the use of TBA as a solvent for the in-vial deposition of 7 (dimethylaminomethylene) amino-9a-methoxymitosane in sterile unit dosage form. This compound is not stable in water. It is introduced into a sterile vial in a TBA solution. Then the TBA is removed by lyophilization. The deposited material contains up to 0.5 mole equivalent of TBA as a hemi-solvate and is very stable to heat.

Miscellaneous applications. United States Surgical Corporation patented a process for preparing foamed, bioabsorbable polymer particles by freeze drying. The particles are useful in medical diagnostic procedures such as mammography and in the repair of damaged or defective bone. The use of TBA or other organic solvents enables the manufacturing process to achieve low processing temperatures that allow medicinals, drugs, growth factors, radiopaque substances, and other additives to be incorporated into the foamed polymer. These additives cannot tolerate high processing temperatures. The bioabsorbable polymer particles serve as excellent vehicles for the delivery of drugs, growth factors, and other biologically active substances to surrounding bone or tissue.

Sterling Drug Inc. patented TBA as a drug dispersion medium for surface modified drug nanoparticles. They claim the use of TBA as a dispersion medium for pharmaceutical drugs having a water solubility of less than 10 mg/mL. The excellent dispersion provides pharmaceutical compositions with unexpectedly high bioavailability.

DeLuca reported that a macromonomer solution with TBA was easier to sterilize by filtration and fill since it was free of foaming compared to the water solution.

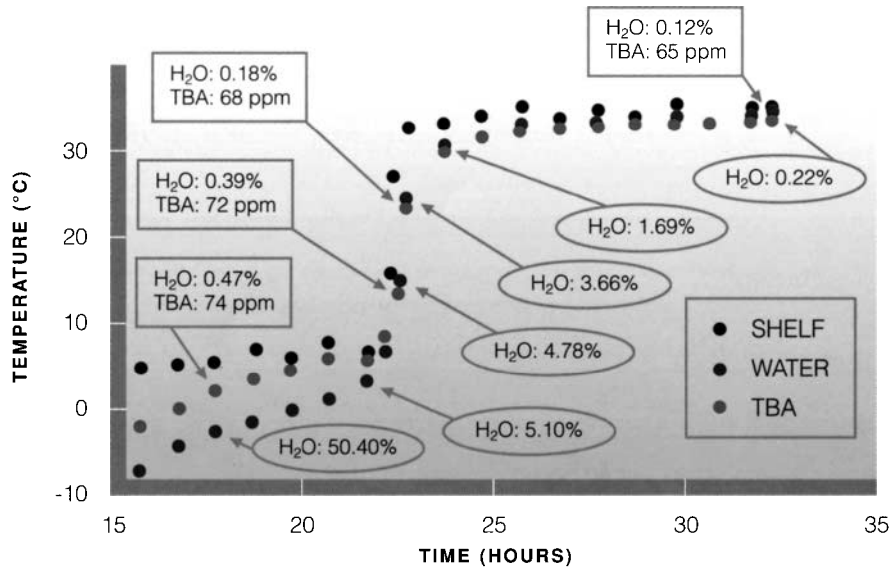


FIG. D-12 Temperature-time profile for freeze-drying cycle of macromonomer. (Source: P. DeLuca, PharmTech Conference Proceeding, 1994, p. 375. Copyright by Advanstar Communications, Inc.)

Figure D-12 illustrates the temperature profile for the samples and the water content at various stages of drying. During the primary drying stage, the TBA solution remained at a lower temperature showing faster drying and the temperature increased after 13 hours showing evidence for lower water content. After 17.5 hours of cycle time, the TBA solution sample reached 1 °C while the water sample remained at 4 °C. The freeze-dried material with TBA showed very low moisture content (0.12 percent) compared to the material freeze dried in water that showed 0.22 percent moisture. The residual TBA was 65 ppm.

Schott Glaswerke has a patent on using TBA to prepare a high purity glass powder with a mean particle size of less than 10 μm. Glass powders having a particle size up to 300 μm are ground to the desired particle size in the presence of a grinding liquid comprising water and TBA. The slurry is then frozen, and the solvent is subsequently removed from the frozen slurry by freeze drying. The resultant glass powder is particularly suitable as a filler for synthetic resins in the dental sector.

Ducting; Ducting and Joints (*see also* Expansion Joints)

Ducting, such as that provided with another major accessory—a gas turbine intake filter system, for example—may be provided by the vendor of the major accessory. If it refers to the gas passageway from the exhaust end of a gas turbine to an HRSG (*see* Cogeneration), the entire package is likely to be provided by the gas turbine vendor. At any rate, ducting of major consequence is generally custom designed for a plant. If well designed in terms of supports and seals and if not subject to fluctuating temperatures, it could well remain a low-maintenance item through the life of a plant. Expansion joints, however, are often subjected to fluctuating temperatures.